



# Testing for herding in the cryptocurrency market

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## ARTICLE INFO

### JEL classification:

G14  
G15

### Keywords:

Behavioral finance  
Herd behavior  
Cryptocurrencies  
Bitcoin

## ABSTRACT

The study investigates whether herding behavior is present in the rapidly emerging cryptocurrency market. By analyzing daily data from major cryptocurrencies during the period August 2015 to December 2018, we find evidence that investors in the cryptocurrency market act irrationally and imitate other's decisions with no reference to their own beliefs. Furthermore, our empirical results provide evidence that the up-events market dispersion follows market movements at a faster pace compared to the down events. Thus, cryptocurrencies show a behavior where they tend to move in tandem, which does not necessarily reflect their fundamentals.

## 1. Introduction

The term cryptocurrency is today used to denote a medium of exchange that is based on cryptography to secure ownership, transactions and the creation of new units of the currency. Today, cryptocurrencies are considered a form of alternative currency and a subset of digital currencies. The way cryptocurrencies are designed and used introduces a paradigm shift in the way ownership, transactions and the money creation process are facilitated. Compared to traditional fiat currencies, cryptocurrencies do not require third parties to address trust issues. Their decentralized nature allows trust issues to be addressed via specialized algorithms. Since their first appearance in 2009, when the Bitcoin cryptocurrency paper was published (Nakamoto, 2009), over 2000 cryptocurrencies have been released and are currently available for trade in international markets. Recently, authors have tried to analyze cryptocurrencies from different perspectives due to the beyond-regular performance that they tend to exhibit (Feng et al., 2017; Katsiampa, 2017). In their analysis, Corbet et al. (2019), state that the majority of studies conducted thus far on cryptocurrencies, have been mainly focusing on market efficiency (Urquhart, 2016; Vidal-Tomás and Ibáñez, 2018; Wei, 2018) and asset pricing bubbles (Cheah and Fry, 2015; Corbet et al., 2017). However, we can cite only a few papers that can be related to examining the behavior of cryptocurrencies. Corbet et al. (2018) examined the reaction of a broad set of digital assets to US Federal Fund interest rate and Quantitative Easing announcements, whereas Kristoufek (2013) examined the relationship between Bitcoin and search queries on Google Trends and Wikipedia. Furthermore, the study of Bouri et al. (2018) produced evidence for a co-explosive behavior in 7 major cryptocurrencies. This interesting finding could be attributed to herding, but also to other potential phenomena, however their analysis has not directly touched upon that issue and remains a matter to be investigated by future literature.

According to the Efficient Market Hypothesis (EMH) contingencies, stocks and other commodities always tend to be traded in their fair prices. However, it is not uncommon that in several instances, capital markets often conflict with the rationality assumptions of the EMH. For instance, based on the results of Bikchandani and Sharma (2001), herding behavior has the tendency to describe a group of individuals acting irrationally and imitating other's decisions with no interest in their own beliefs.

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<https://doi.org/10.1016/j.frl.2019.06.008>

Received 15 February 2019; Received in revised form 15 May 2019; Accepted 21 June 2019

Available online 22 June 2019

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**Table 1**Estimation results of  $CSAD_t$ ,  $CSAD_t^{UP}$  and  $CSAD_t^{DOWN}$  using Newey–West standard errors.

| $CSAD_t = a + \gamma_1  R_{m,t}  + \gamma_2 R_{m,t}^2 + \varepsilon_t$ |                      |                     | $CSAD_t^{UP} = a + \gamma_1^{UP}  R_{m,t}^{UP}  + \gamma_2^{UP} (R_{m,t}^{UP})^2 + \varepsilon_t$ |                      |                    | $CSAD_t^{DOWN} = a + \gamma_1^{DOWN}  R_{m,t}^{DOWN}  + \gamma_2^{DOWN} (R_{m,t}^{DOWN})^2 + \varepsilon_t$ |                      |                    |
|------------------------------------------------------------------------|----------------------|---------------------|---------------------------------------------------------------------------------------------------|----------------------|--------------------|-------------------------------------------------------------------------------------------------------------|----------------------|--------------------|
| $\alpha$                                                               | $\gamma_1$           | $\gamma_2$          | $\alpha$                                                                                          | $\gamma_1^{UP}$      | $\gamma_2^{UP}$    | $\alpha$                                                                                                    | $\gamma_1^{DOWN}$    | $\gamma_2^{DOWN}$  |
| 0.0273<br>(0.000)***                                                   | 0.6749<br>(0.000)*** | -1.7385<br>(0.057)* | 0.0279<br>(0.000)***                                                                              | 0.7721<br>(0.000)*** | -1.1845<br>(0.418) | 0.0276<br>(0.000)***                                                                                        | 0.4554<br>(0.001)*** | -1.1861<br>(0.219) |
| Obs                                                                    | F (2, 1239)          | Prob > F            | Obs                                                                                               | F (2, 669)           | Prob > F           | Obs                                                                                                         | F (2, 567)           | Prob > F           |
| 1242                                                                   | 23.61                | 0.0000              | 672                                                                                               | 30.64                | 0.0000             | 570                                                                                                         | 8.60                 | 0.0002             |

(\*\*\*), (\*\*), (\*) denotes that the coefficient is significant at the (1%), (5%), (10%) level.

Christie and Huang (1995) suggest that investors depend on overall market conditions regarding their decision process. During normal periods individual investors tend to trade based on their own private information, a phenomenon offering diversity to the final outcome. However, during periods of extreme and irregular market movements, individuals tend to suppress their own rational thinking and are more likely to follow market's reactions.

The attempt of economists and investors to understand herding behavior has received considerable attention over recent years. In their studies Christie and Huang (1995); Chang et al. (2000); Gleason et al. (2004) have rejected the null hypothesis of herd behavior existing in the US stock market. However, Chiang and Zheng (2010) and Khan et al. (2011) have provided evidence in favor of the presence of herd behavior in various developed European countries. Furthermore, the results of Chiang and Zheng (2010) have stood in contrast to earlier literature that has shown no herding in advanced markets (Chang et al., 2000) and in Chinese markets (Demirer and Kutan, 2006).

In this study our aim is to contribute to the expanding cryptocurrency literature, by investigating whether herding behavior is present in cryptocurrencies markets. Up until now, there are only two other studies examining aspects of this issue. Bouri et al. (2018) used a rolling window analysis to highlight that herding varies over time in the market,<sup>1</sup> and Vidal-Tomás et al. (2018) observed in their analysis of 65 digital currencies, that the smallest cryptocurrencies are herding with the largest ones. Considering this theoretical framework, this letter will contribute to the literature by examining whether herding behavior is present among the top players of the cryptocurrency market. The criteria that we have used in order to define the “top” players under examination are: (i) Market capitalization, (ii) Market volume, (iii) Availability of data for the entire horizon of our estimation and (iv) Cryptocurrencies under examination should be listed for trading in major cryptocurrency platforms. Thus, our empirical tactic will utilize the “top” 6 cryptocurrencies, under the criteria mentioned earlier, over a 3-year period and investigate whether they exhibit herding behavior. Although, an intrinsic characteristic of cryptocurrencies trading is that they are traded in decentralized electronic platforms, still the information on all cryptocurrencies is available in real time to all involved parties at the palm of their hand, creating a digital global investment community. This real time availability of information may lead to herding consistent trading decisions.

## 2. Data issues and empirical methodology

Since herding behavior is likely to manifest in high frequency data (Christie and Huang, 1995), we utilize daily prices for 6 major cryptocurrencies,<sup>2</sup> that represent, in average, over 85%<sup>3</sup> of market capitalization between August 2015 to December 2018 generating 1242 observations. The cryptocurrencies under study are: (i) Bitcoin, (ii) Dash, (iii) Ethereum, (iv) Litecoin, (v) Monero and (vi) Ripple.<sup>4</sup>

For each cryptocurrency we calculate the daily return:

$$R_{i,t} = \ln \frac{P_{i,t}}{P_{i,t-1}} \quad (1)$$

Where  $i$  denotes the cryptocurrency,  $t$  the time period, and  $P$  the closing price for each cryptocurrency.

Christie and Huang (1995) suggested a metric to capture the dispersion of beliefs across assets, by using the cross-sectional standard deviation (CSSD) method:

$$CSSD_t = \sqrt{\frac{\sum_{i=1}^N (R_{i,t} - R_{m,t})^2}{N-1}} \quad (2)$$

where  $R_{i,t}$  is the observed stock return of firm  $i$  at time  $t$ ,  $R_{m,t}$  is the cross-sectional average return of the  $N$  returns in the equally-weighted market portfolio at time  $t$ , and  $N$  is the number of stocks in the market portfolio.

<sup>1</sup> Bouri et al. (2018) use 14 cryptocurrencies and a period analysis from 2013–2018, however it should be noted that the count of digital currencies varies significantly during the period covered. For instance, it is only in the post 2016 period that all 14 cryptocurrencies are traded.

<sup>2</sup> <https://www.cryptocompare.com>.

<sup>3</sup> <https://www.coinmarketcap.com>.

<sup>4</sup> In our analysis we have employed cryptocurrencies that have been trading the entire sample period in major cryptocurrency platforms and had a significant size of both market capitalization and trade volume.

**Table 2**  
Estimation results of  $CSAD_b$ ,  $CSAD_t^{UP}$  and  $CSAD_t^{DOWN}$  for the GARCH model.

| $a$                                 | $CSAD_t = a + \gamma_1  R_{m,t}  + \gamma_2 R_{m,t}^2 + \varepsilon_t$ | $\gamma_2$                         | $CSAD_t^{UP} = a + \gamma_1^{UP}  R_{m,t}  + \gamma_2^{UP} (R_{m,t}^{UP})^2 + \varepsilon_t$ | $\gamma_1^{UP}$      | $\gamma_2^{UP}$                       | $CSAD_t^{DOWN} = a + \gamma_1^{DOWN}  R_{m,t}  + \gamma_2^{DOWN} (R_{m,t}^{DOWN})^2 + \varepsilon_t$ | $\gamma_1^{DOWN}$     | $\gamma_2^{DOWN}$                   |
|-------------------------------------|------------------------------------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------|----------------------|---------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------|-------------------------------------|
| 0.0185<br>(0.000)***<br>ARCH(lag 1) | 0.2161<br>(0.000)***                                                   | -0.1724<br>(0.542)<br>GARCH(lag 1) | 0.2511<br>(0.000)***<br>ARCH(lag 1)                                                          | 0.9074<br>(0.000)*** | -2.9377<br>(0.000)***<br>GARCH(lag 1) | 0.2754<br>(0.000)***<br>ARCH(lag 1)                                                                  | 0.4715<br>(0.000)***  | -1.3605<br>(0.053)*<br>GARCH(lag 1) |
| 0.2331<br>(0.000)***                | 0.7948<br>(0.000)***                                                   | 0.0000<br>(0.000)***               | 0.2772<br>(0.000)***                                                                         | 0.7182<br>(0.000)*** | 0.7182<br>(0.799)                     | -0.0010<br>(0.000)***                                                                                | -1.2630<br>(0.000)*** | -1.2630<br>(0.000)***               |
| Obs                                 | Wald chi2(2)                                                           | Prob > chi2                        | Obs                                                                                          | Wald chi2(2)         | Prob > chi2                           | Obs                                                                                                  | Wald chi2(2)          | Prob > chi2                         |
| 1242                                | 171.08                                                                 | 0.0000                             | 672                                                                                          | 148.87               | 0.0000                                | 570                                                                                                  | 27.28                 | 0.0000                              |

(\*\*\*), (\*\*), (\*) denotes that the coefficient is significant at the (1%), (5%), (10%) level.

In a later study [Chang et al. \(2000\)](#) have suggested the use of the cross-sectional absolute deviation (CSAD), since CSSD tends to be sensitive to outliers, due to the fact that is calculated by squared return-deviations. The cross-sectional absolute deviation (CSAD) is expressed as:

$$CSAD_t = \frac{1}{N} \sum_{i=1}^N |R_{i,t} - R_{m,t}| \quad (3)$$

The empirical investigation of herding was conducted by [Chang et al. \(2000\)](#), as in the following regression model:

$$CSAD_t = a + \gamma_1 |R_{m,t}| + \gamma_2 R_{m,t}^2 + \varepsilon_t \quad (4)$$

where  $|R_{m,t}|$  is the absolute equally-weighted market return and  $R_{m,t}^2$  is the squared market return.

The relationship formed between  $CSAD_t$  and  $R_{m,t}$  is used so that herd behavior to be detected. According to [Chang et al. \(2000\)](#), a statistically significant negative coefficient  $\gamma_2$  implies the presence of herd behavior. Such an event is likely to increase the correlation between individual asset returns, and the dispersion between asset returns will either decrease or increase at a decreasing rate. In periods described by large price volatility, if investors are more susceptible to herding, then there should be a less than proportionate increase/decrease in the CSAD measure. If herding behavior does not exist, the dispersion increases in proportion with the increasing returns of the market.

Furthermore, in the relationship assumed between CSAD and market returns, asymmetry may exist ([Chang et al., 2000](#)). This hypothesis can be tested with the estimation of two different models:

$$CSAD_t^{UP} = a + \gamma_1^{UP} |R_{m,t}^{UP}| + \gamma_2^{UP} (R_{m,t}^{UP})^2 + \varepsilon_t, \text{ if } R_{m,t} > 0 \quad (5)$$

$$CSAD_t^{DOWN} = a + \gamma_1^{DOWN} |R_{m,t}^{DOWN}| + \gamma_2^{DOWN} (R_{m,t}^{DOWN})^2 + \varepsilon_t, \text{ if } R_{m,t} < 0 \quad (6)$$

where  $|R_{m,t}^{UP}|$  ( $|R_{m,t}^{DOWN}|$ ) is the equally-weighted average return of the  $N$  stocks available on day  $t$  when the return is positive (negative). Accordingly,  $CSAD_t^{UP}$  is the  $CSAD_t$  for the day  $t$  where  $R_{m,t}$  is positive and  $CSAD_t^{DOWN}$  is the  $CSAD_t$  for the day  $t$  where  $R_{m,t}$  is negative.

### 3. Empirical results

Eq. (4) was used to assess whether the behavior of investors in the cryptocurrency market is compatible with herding or not. A negative and statistically significant value of the  $R_{m,t}^2$  coefficient indicates the existence of herding in the market. Eqs. (5) and (6) were accordingly used to examine the possibility that herding behaves asymmetrically across upward and downward market movements. [Chang et al. \(2000\)](#) and [Mobarek et al. \(2014\)](#) suggest that the standard errors of the estimated regressions should be adjusted for heteroscedasticity and autocorrelation using the method proposed by [Newey and West \(1987\)](#). Therefore, in order to address this issue, the [Newey and West \(1987\)](#) estimator is used in this analysis. We also offer, in Table 2, a Generalized Autoregressive Conditional Heteroskedasticity model and more specifically a GARCH (1,1) model<sup>5</sup> [Bollerslev et al., 1992](#) for all specifications. These estimations are an attempt to capture the sample properties of data with respect to heteroskedasticity. Tables 1 and 2 report the results for models (4), (5) and (6) using the OLS estimator using Newey–West standard errors and the GARCH model respectively.

In the first specification where  $CSAD_t$  is the dependent variable, we find that the coefficient  $\gamma_1$  is positive and statistically significant at the 1% level in both the Newey–West estimation and the GARCH model. Moving now to the main parameter of interest,  $\gamma_2$ , it is found to be negative ( $-1.7385$ ) and statistically significant (at the 5.7% significance level) in the Newey–West estimation (Table 1) and negative, albeit not statistically significant, in the GARCH model (Table 2). This finding is compatible with herding behavior since it suggests that market dispersion movements are less than proportionate to market returns movements. Furthermore, it is in contrast to previous findings by [Bouri et al. \(2018\)](#) and [Vidal-Tomás et al. \(2018\)](#) who have stated in their studies that there is no evidence of herding in the static approach of the generalized form.

The second and third specifications, not only investigate herding in the decomposed  $CSAD_t$  between upward and downward, but also focus on the possibility of an asymmetric reaction between the ups and downs. In terms of herding, in the Newey–West estimation (Table 1) for the second and third specifications both  $\gamma_2$  coefficients are negative  $\gamma_2^{UP}$  ( $-1.1845$ ),  $\gamma_2^{DOWN}$  ( $-1.1861$ ), however the results are found to be statistically insignificant. In the GARCH model (Table 2) our results support herding presence since both  $\gamma_2$  coefficients are significantly negative. A comparison of the two-point estimates, in the GARCH model, indicates that  $\gamma_2$  coefficient in the up model is  $-2.9377$  while in the down model is  $-1.3605$ , findings which provide informal evidence that the up-events market dispersion follows market movements at a faster pace compared to the down events. In reference to related literature, [Vidal-Tomás et al. \(2018\)](#) have also observed in their study the presence of asymmetric herding behavior, however only during down movements of the cryptocurrency market.

### 4. Conclusion

The aim of this study was to explore whether herding behavior exists in the cryptocurrency market. The analysis was based on daily data from a sample of major cryptocurrencies that represent, in average, over 85% of total market capitalization covering the period August 2015 to December 2018.

<sup>5</sup> The Bayesian information criterion (BIC), developed by [Schwarz \(1978\)](#), was used for the model selection.

Our results indicate that herding is indeed present. By using the cross-sectional absolute deviation model, the current study uncovered a statistically significant negative  $\gamma_2$  ( $-1.7385$ ) coefficient indicating the presence of herding behavior in the market. This suggests that as the average equally-weighted market return increases in absolute terms, the cross-sectional absolute deviation increases, but at a decreasing rate. In addition, based on the GARCH model, we find that during up movements the previously mentioned decreasing rate of reaction is even more pronounced  $\gamma_2^{UP}$  ( $-2.9377$ ).

Overall, our findings suggest that investors in the “top” sector of the cryptocurrency market act irrationally and imitate other's decisions with no reference to their own beliefs. Thus, cryptocurrencies show a behavior where they tend to move in tandem, which does not necessarily reflect their fundamentals. This herding behavior could be a partial explanation of the co-explosivity findings reported in the literature, but also could generate a co-shrinkage of cryptocurrencies.

## Acknowledgments

The authors would like to thank the Editor and the Referees for their comments that improved the paper. This research is co-financed by Greece and the European Union (European Social Fund-ESF) through the Operational Programme «Human Resources Development, Education and Lifelong Learning» in the context of the project “Strengthening Human Resources Research Potential via Doctorate Research” (MIS-5000432), implemented by the State Scholarships Foundation (IKY).

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